Sonia Nath

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EDUCATION

Portland State University **Expected Graduation 2027**
Bachelor of Science in Computer Science *Portland, OR*

- GPA: 3.94; President's List
- Relevant Coursework: Data Structures, System Programming & Architecture, Calculus I-III, Discrete Structures I-II, Algorithms & Complexity, Operating Systems, Database Management Systems, Artificial Intelligence, Data Engineering, Internet & Web & Cloud Systems
- Organizations: Louis Stokes Alliances for Minority Participation (LSAMP), Uplift

SKILLS

- **Programming & Scripting:** Python, SQL, C/C++, x86 Assembler, JavaScript/HTML/CSS
- **AI & Machine Learning:** Retrieval-Augmented Generation (RAG), OpenAI API, FAISS, Prompt Engineering
- **Data Engineering:** Databricks, Apache Spark, pandas, ETL Pipelines, Workday API Integration, Power BI
- **Developer Tools:** Github/Git, Vim
- **Frameworks:** Flask

WORK EXPERIENCE

Data Engineer Intern – Data & Business Intelligence **Jan. 2026 – July 2026**
The Standard *Portland, OR*

- Built an end to end data pipeline in **Databricks** to automate ingestion of Workday employee, contractor and time tracking data, replacing a manual Excel based reporting process.
- Developed ETL workflows using **Python** (**pandas/Spark**) and **SQL** to clean, transform, and model data into **Power BI** ready tables and unified workforce datasets.
- Integrated historical Excel data with live **Workday API** feeds and enabled automated Power BI reporting through scheduled jobs, eliminating manual data refreshes and improving data consistency.

Software Engineering Intern **June 2025 – Sept. 2025**
CDK Global *Portland, OR*

- Developed an AI-powered coding assistant using **Retrieval-Augmented Generation** (RAG) to improve code generation accuracy within the Proton Blueprint Editor.
- Designed and optimized a **FAISS** vector database for semantic retrieval by embedding internal plugin documentation, enabling low-latency retrieval of relevant context for LLM responses.
- Built a **Flask** backend with asynchronous request handling, secure embedding workflows, inline autocomplete, multi-suggestion cycling and code block insertion features.

Undergraduate Researcher **Sept. 2023 – May 2025**
Teuscher:.lab - LSAMP Research Scholars Program (2024 – 2025) *Portland, OR*

- Designed and implemented Spiking Neural Network (SNN) and Liquid State Machine (LSM) models using **snnTorch** and **NumPy** to solve grid based binary classification task.
- Utilized Spike Timing Dependent Plasticity (STDP) rule to tune LSM to criticality, investigating the impact of different learning types on the network's ability to recognize dynamic patterns.
- Documented methodologies, experimental results and insights as part of ongoing research on optimizing efficient spiking neural networks and advancing neuromorphic computing systems.

Wireless Environmental Sensor Technology Lab - LSAMP Research Scholars Program (2023 – 2024)

- Led an independent project using **Python** to design accessible software for streamlining the extraction of temperature data from iButton devices via the one-wire communication protocol.
- Gained hands on experience interfacing hardware (iButton and microcontrollers) with **Python** based software, overcoming various technical challenges.
- Presented project findings at the LSAMP Pacific Northwest Research Conference, reaching an audience of 100+.